Lecture 9

Approximating Solutions to IVPs

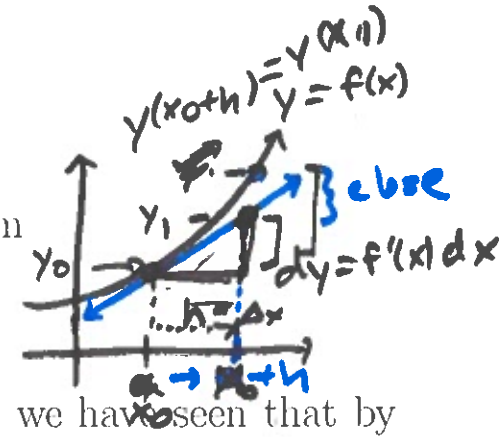
We have learned various methods to solve 1st-order initial value problems of the form

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

However, not every equation of this form has a simple solution technique. In fact, not every equation of this form may have an explicit particular solution.

In these circumstances, we can look to find certain values of the solution through approximations.

Euler's Method



Suppose we are given the 1st-order initial value problem

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

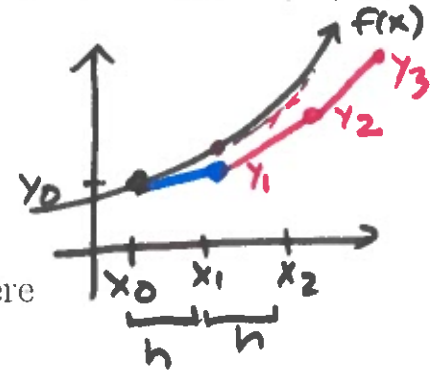
in the study of differentials and linear approximations we have seen that by replacing $\frac{dy}{dx}$ by $\frac{\Delta y}{\Delta x}$ we can obtain an approximation for a future value of y given the point (x_0, y_0).

$$\frac{\Delta y}{\Delta x} = f(x,y) \Rightarrow \Delta y \approx \Delta x\, f(x,y)$$

Writing $\Delta y = y_1 - y_0$ and $\Delta x = x_1 - x_0 = h$ for a fixed h called the **step size** we can rewrite our IVP as

$$\frac{y_1 - y_0}{h} = f(x_0, y_0) \quad \Rightarrow \quad y_1 - y_0 = hf(x_0, y_0) \quad \Rightarrow \boxed{y_1 = \overbrace{y_0}^{\text{rise}} + \underbrace{h}_{\text{run}}\, \underbrace{f(x_0, y_0)}_{\text{slope} = \frac{\text{rise}}{\text{run}}}}$$

This given an approximation for the solution y at x_1 by the coordinate (x_1, y_1), that is $y(x_1) \approx y_1$. We can repeat this process to obtain another value $y(x_2) =\approx y_2$, etc. This process is called **Euler's Method**.



Theorem: Given the IVP

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

we can approximate $y(x_i) \approx y_i$ using Euler's Method where

$$\begin{cases} x_{i+1} = x_i + h \\ y_{i+1} = y_i + hf(x_i, y_i) \end{cases}$$

using a prescribed step size h.

Example: Use Euler's Method with step size $h = 0.25$ to approximate the solution to

$$\frac{dy}{dx} = \underbrace{x + 2y}_{f(x,y)}, \quad y(\overset{x_0}{0}) = \underset{y_0}{0} \qquad (x_0 = 0, y_0 = 0)$$

at $x = 0.75$ (3 steps)

$h = 0.25$

$$0 \xrightarrow{h} 0.25 \xrightarrow{h} 0.5 \xrightarrow{h} 0.75$$

$$y_0 \to y_1 \to y_2 \to y_3$$

$$\begin{cases} x_1 = x_0 + h = 0 + 0.25 = 0.25 \\ y_1 = y_0 + h f(x_0, y_0) = 0 + 0.25 f(0,0) = 0 + 0.25(0 + 2(0)) \\ \qquad = 0 \end{cases}$$

New pt $(x_1 = 0.25, y_1 = 0)$

$$\begin{cases} x_2 = x_1 + h = 0.25 + 0.25 = 0.5 \\ y_2 = y_1 + h f(x_1, y_1) = 0 + 0.25(f(0.25, 0)) = 0 + 0.25(0.25 + 2(0)) \\ \qquad = 0.0625 \end{cases}$$

new pt $(x_2 = 0.5, y_2 = 0.0625$

$$\begin{cases} x_3 = x_2 + h = 0.5 + 0.25 = 0.75 \\ y_3 = y_2 + hf(x_2, y_2) = 0.0625 + 0.25(f(0.5, 0.0625)) \\ \qquad = 0.0625 + 0.25(0.5 + 2(0.0625)) \\ \qquad = 0.21875 \end{cases}$$

We have that $\underset{\text{true value}}{y(0.75)} \approx y_3 = 0.21875$

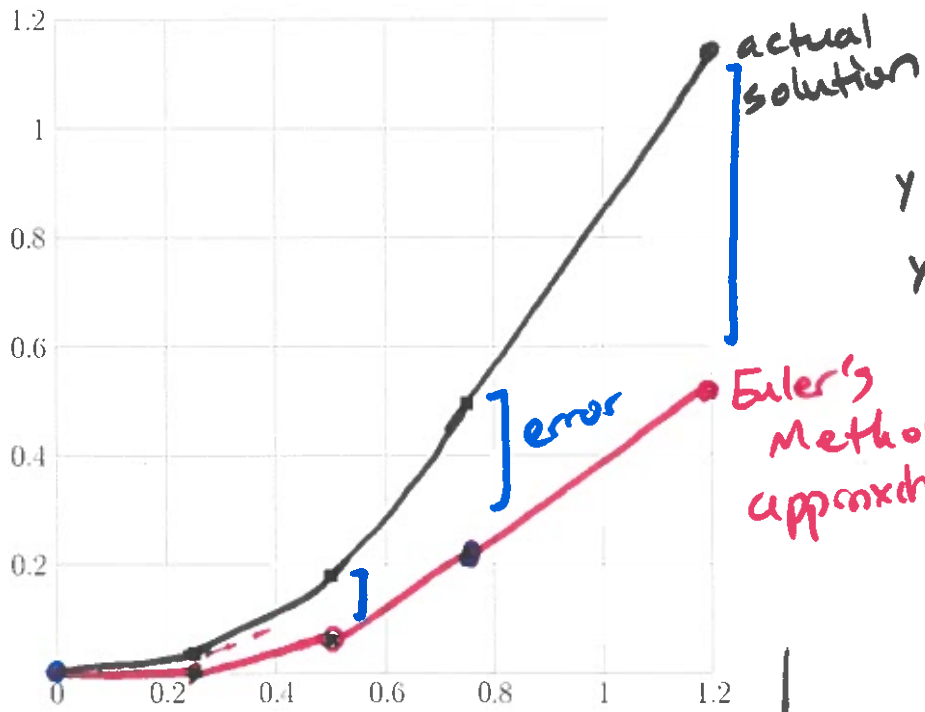
divide step size by 10.

This approximating technique is not very accurate (gain ≈ 1 decimal accuracy per step). We can solve the previous IVP exactly (it is linear) and obtain

$\frac{dy}{dx} - 2y = x, \; y(0) = 0$

$$y = 0.25e^{2x} - 0.5x - 0.25$$

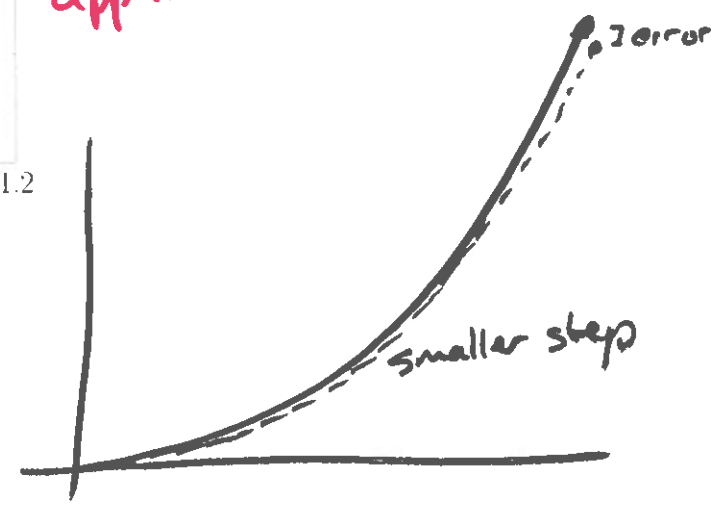
We can plot the graph of the solution and approximating values below. The values are given in the table:



$y' = 0.5e^{2x} - 0.5$

$y'(0) = 0.5 - 0.5 = 0$

x	y (Euler)	y (Actual)
0	0	0
0.25	0	0.03718
0.5	0.0625	0.17957
0.75	0.21875	0.495422



There are two ways to increase the accuracy of the approximation. We can reduce the step size. If we were to use $h = 0.025$ in the previous problem it would take 30 steps to get from $x = 0$ to $x = 0.75$ instead of 3 steps. However, our approximation would be $y(0.75) \approx \underbrace{0.455486}_{y_{30}}$ which is much closer to the actual value of 0.495422.

.455486 closer to 0.495422

The other way to increase the accuracy is the use a more accurate approximation technique.

Taylor Methods

Along the same lines as $\overbrace{\text{Euler's Method}}^{\text{linear}}$, suppose that y is the solution to the initial value problem

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0.$$

If we expand y using a Taylor Series centered at x_0 we obtain

$$y(x) = y(x_0) + y'(x_0)(\underbrace{x - x_0}_{h}) + \frac{y''(x_0)}{2!}(\underbrace{x - x_0}_{h})^2 + \frac{y'''(x_0)}{3!}(\underbrace{x - x_0}_{h})^3 + \cdots$$

We can substitute $x = x_1$, $x_1 - x_0 = h$, $y(x_0) = y_0$, and $y'(x_0) = \frac{dy}{dx}(x_0) = f(x_0, y_0)$ and write

$$y(x_1) = \underbrace{y_0 + hf(x_0, y_0)}_{\text{Euler's Method}} + \frac{h^2}{2!}y''(x_0) + \frac{h^3}{3!}y'''(x_0) + \cdots$$

Therefore, if we want y_1 from Euler's Method we truncate this expression after the linear term and remove $\frac{h^2}{2!}y''(x_0) + \frac{h^3}{3!}y'''(x_0) + \cdots$.

The true value of $y(x_{i+1})$ in Euler's method is

$$y(x_{i+1}) = y_i + hf(x_i, y_i) + \frac{h^2}{2!}y''(x_i) + \frac{h^3}{3!}y'''(x_i) + \cdots$$

and so the error in Euler's Method is exactly

$$y(x_{i+1}) - y_{i+1} = \left(y_i + hf(x_i, y_i) + \frac{h^2}{2!}y''(x_i) + \frac{h^3}{3!}y'''(x_i) + \cdots \right) - \underbrace{(y_i + hf(x_i, y_i))}_{\text{Euler's}}$$

$$= \frac{h^2}{2!}y''(x_i) + \frac{h^3}{3!}y'''(x_i) + \cdots \quad = h^2(\sim)$$

Since the error term depends upon h^2 we say that the error in Euler's Method is $O(h)$ and the method is a 1st-order method.

→ method stops at h^1

If we instead take more terms of the Taylor expansion we can get higher-order methods. In order to do so we must express $y^{(n)}(x)$ in terms of $f(x, y)$ and its partial derivatives. As $y' = f(x, y)$ we can differentiate and obtain

$$y'' = \frac{d}{dx}(f(x, y)) = \frac{\partial f}{\partial x}(x, y) + \frac{\partial f}{\partial y}(x, y) \cdot \frac{dy}{dx} = \frac{\partial f}{\partial x}(x, y) + \frac{\partial f}{\partial y}(x, y) \cdot f(x, y)$$

So we define $f_2(x, y) = \frac{\partial f}{\partial x}(x, y) + \frac{\partial f}{\partial y}(x, y) \cdot f(x, y)$. In a similar fashion we can continue to differentiate to obtain expressions for y''', $y^{(4)}$....

$y''' = f_3(x,y)$, $y^{(4)} = f_4(x,y)$

Theorem (Taylor Methods of Order n)**:** Given the IVP

$$\frac{dy}{dx} = f(x, y), \quad f(x_0) = y_0$$

The **Taylor Method of Order n** with step size h is given by the recursive formula

$$\begin{cases} x_{i+1} = x_i + h \\ y_{i+1} = \underbrace{y_i + hf(x_i, y_i)}_{\text{Euler}} + \frac{h^2}{2!}f_2(x_i, y_i) + \cdots + \frac{h^n}{n!}f_n(x_i, y_i) \end{cases}$$

Example: Use the Taylor Method of Order 2 with step size $h = 0.5$ to approximate the solution to

$$\frac{dy}{dx} = \underbrace{xy - y^2}_{f(x,y)=xy-y^2}, \quad y(\underset{x_0}{0}) = \underset{y_0}{-1} \qquad (x_0=0,\ y_0=-1$$

→ no explicit solution

at $x = 1$.

$0 \xrightarrow[h]{} 0.5 \xrightarrow[h]{} 1$

The Method is order 2.

$$\begin{cases} x_{i+1} = x_i + h \\ y_{i+1} = y_i + hf(x_i, y_i) + \frac{h^2}{2} f_2(x_i, y_i) \end{cases}$$

$$= \begin{cases} x_{i+1} = x_i + h \\ y_{i+1} = y_i + hf(x_i, y_i) + \frac{h^2}{2}\left[\frac{\partial f}{\partial x}(x_i, y_i) + \frac{\partial f}{\partial y}(x_i, y_i) f(x_i, y_i)\right] \end{cases}$$

$x_0 = 0,\ y_0 = -1,\ h = 0.5$

$$\begin{cases} x_1 = x_0 + 0.5 = 0.5 \\ y_1 = y_0 + hf(x_0, y_0) + \frac{h^2}{2}\left[f_x(x_0, y_0) + f_y(x_0, y_0) f(x_0, y_0)\right] \end{cases}$$

Note $f_x = y$, $f_y = x - 2y$

so $y_1 = -1 + 0.5 f(0,-1) + \frac{0.5^2}{2}\left[f_x(0,-1) + f_y(0,-1) f(0,-1)\right]$

$= -1 + 0.5(-1) + \frac{0.25}{2}\left[-1 + 2(-1)\right]$

$= -1 - 0.5 + 0.125(-3) = -1.875$

$x_1 = 0.5,\ y_1 = -1.875$ $\qquad f(x,y) = xy - y^2$

$x_2 = 0.5 + 0.5 = 1$

$y_2 = y_1 + hf(x_1, y_1) + \frac{h^2}{2}\left[f_x(x_1, y_1) + f_y(x_1, y_1) f(x_1, y_1)\right]$

$= -1.875 + 0.5(0.5(-1.875) - (-1.875)^2)$

$+ \frac{0.5^2}{2}\left[-1.875 + (0.5 - 2(-1.875)) \cdot (0.5(-1.875) - (-1.875)^2)\right]$

$= -6.70166\ 015625$

Bernoulli: $y(1) \underset{\text{true}}{\approx} -8.45$

Runge-Kutta Methods

Euler's Method had error terms depending upon h^2 and so it is a 1st-order method. One way to express this is to write

$$y(x_{i+1}) = y_i + hf(x_i, y_i) + O(h^2)$$

The Taylor Method of Order n has error terms depending upon h^{n+1} and is an nth-order method and the convergence is $O(h^n)$ and so we would have $O(h^{n+1})$ in the expansion of the true value of y.

$f_2, f_3, f_4, \ldots$

One of the issues with using a Taylor Method of higher order is the computation of the expressions for each derivative in terms of $f(x, y)$ and its partial derivatives. There are techniques and methods that help circumvent this requirement. One is the **Runge-Kutta** methods.

The idea behind a Runge-Kutta method is to write the terms after y_i in the Taylor Methods as $\boxed{k_n = hf(x, y, h, k_1, k_2, \ldots, k_{n-1})}$ which does not rely on computing any partial derivatives of $f(x, y)$.

$y_{i+1} = y_i + k_1$ nth-order method

Write the Taylor Method of Order 2 as

$$y(x_{i+1}) = y_i + hT_2 + O(h^3) = y_i + hf(x_i, y_i) + \frac{h^2}{2}[f_2(x_i, y_i)] + O(h^3)$$

where

$$T_2 = f(x, y) + \frac{h}{2}\left[\frac{\partial f}{\partial x}(x, y) + \frac{\partial f}{\partial y}f(x, y)\right]$$

with $K_2 = f(x + \alpha h, y + \beta hf(x, y))$ by choosing appropriate values for α and β.

If we expand K_2 about $h = 0$ we have

$$\begin{aligned} K_2(h) &= K_2(0) + \frac{dK_2}{dh}(0)h + O(h^2) \\ &= f(x,y) + \left[\alpha\frac{\partial f}{\partial x}(x,y) + \beta\frac{\partial f}{\partial y}(x,y)f(x,y)\right]h + O(h^2) \end{aligned}$$

Taylor

and by subtracting T_2 and K_2 about $h = 0$ we see $T_2 - K_2$ is equal to

$$\underbrace{\left(\frac{h}{2}\left[\frac{\partial f}{\partial x}(x,y) + \frac{\partial f}{\partial y}f(x,y)\right]\right)}_{T_2} - \left(\underbrace{\left[\alpha\frac{\partial f}{\partial x}(x,y) + \beta\frac{\partial f}{\partial y}(x,y)f(x,y)\right]}_{K_2} h + O(h^2)\right)$$

and note that in order for the convergence to stay $O(h^2)$ we must have this subtraction be zero. This means that $\alpha = \beta = \frac{1}{2}$.

Definition: The **Runge-Kutta Method of Order 2** to solve the IVP

$$\frac{dy}{dx} = f(x,y), \quad y(x_0) = y_0$$

with step size h has recursive formulas

$$\begin{cases} x_{i+1} = x_i + h \\ y_{i+1} = y_i + hf\left(x_i + \frac{h}{2}, y_i + \frac{h}{2}f(x_i, y_i)\right) \end{cases}$$

replaces $f(x_i, y_i) + \frac{h^2}{2}[f_x + f_y f]$

One can extend this to any order by making the same computations to the appropriate order derivatives and comparing with the Taylor methods. The practical method that is relatively easy to implement is the Runge-Kutta Method of Order 4, which is usually called the **classical 4th-order Runge-Kutta Method**.

Definition: Given the IVP

$$\frac{dy}{dx} = f(x,y), \quad y(x_0) = y_0$$

the **classical 4th-order Runge-Kutta Method** is given by the recursive formulas

$$\begin{cases} x_{i+1} = x_i + h \\ y_{i+1} = y_i + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \end{cases}$$

where

no need to memorize.

$$k_1 = hf(x_i, y_i)$$
$$k_2 = hf\left(x_i + \frac{h}{2}, y_i + \frac{k_1}{2}\right)$$
$$k_3 = hf\left(x_i + \frac{h}{2}, y_i + \frac{k_2}{2}\right)$$
$$k_4 = hf(x_i + h, y_i + k_3)$$

Example: Use the classical 4th-order Runge-Kutta method to approximate the solution to

$y = e^x$ $\quad y(1) = e^1 = e \approx 2.7182818$

$$y' = y, \quad y(0) = 1$$

$x_0 = 0$ $\quad y_0 = 1$

at $x = 1$ with step size $h = 1$

$f(x,y) = y$

$0 \xrightarrow[h]{} 1$ one step.

$x_0 = 0, \; y_0 = 1, \; h = 1$

$$\begin{cases} k_1 = hf(x_0, y_0) = 1 \cdot f(0,1) = 1 \\ k_2 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_1}{2}) = 1 \cdot f(0 + \frac{1}{2}, 1 + \frac{1}{2}) \\ \qquad = 1 \cdot f(\frac{1}{2}, \frac{3}{2}) = \frac{3}{2} \\ k_3 = hf(x_0 + \frac{h}{2}, y_0 + \frac{k_2}{2}) = 1 \cdot f(0 + \frac{1}{2}, 1 + \frac{3/2}{2}) \\ \qquad = f(\frac{1}{2}, 7/4) = 7/4 \\ k_4 = hf(x_0 + h, y_0 + k_3) = 1 \cdot f(0+1, 1 + 7/4) = \frac{11}{4} \end{cases}$$

so $y_1 = y_0 + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$

$$\Rightarrow y_1 = 1 + \frac{1}{6}\left(1 + 2\left(\frac{3}{2}\right) + 2\left(7/4\right) + 11/4\right)$$

$$= 1 + \frac{1}{6}\left(1 + 3 + \frac{7}{2} + \frac{11}{4}\right) = 1 + \frac{1}{6}\left(\frac{41}{4}\right)$$

$$= 1 + \frac{41}{24} = 2.7083\overline{3}$$

very close: error $\underset{e}{2.718828 - 2.70833}$

$= 0.09005$

two steps get 4 decimals of accuracy

$h = 0.5$